

Experimental Results

Evaluation of Physics-VAE Synthetic Data Generation Framework

Overview

The proposed Physics-VAE framework was evaluated against three baseline synthetic data generation approaches: Physics-MC, Regression, and VAE. Experiments were conducted on four agricultural datasets representing two different modelling scenarios: quality prediction (Apple and Banana) and nutrient composition modelling (Date and Safou).

The evaluation considers three complementary aspects of synthetic data quality. Statistical fidelity is assessed using correlation distance and the mean Kolmogorov–Smirnov (KS) statistic, calibration quality is measured using calibration error, and downstream usefulness is evaluated using Train-on-Synthetic Test-on-Real (TSTR) accuracy. Lower values are preferred for fidelity and calibration error metrics, whereas higher TSTR accuracy indicates better synthetic data utility.

1. Synthetic Data Generation Performance

Table 1 presents the comparison between the proposed Physics-VAE framework and baseline approaches. The results demonstrate that Physics-VAE improves synthetic data quality by combining physics-based prior information with variational representation learning.

Table 1: Comparison of synthetic data generation methods across quality and nutrient datasets. Lower values are preferred for correlation distance, mean KS statistic, and calibration error, while higher values are preferred for TSTR accuracy. The best-performing synthetic generator is highlighted in bold.

Dataset (Type)	Method	Correlation Distance ↓	Mean KS ↓	Calibration Error ↓	TSTR Accuracy ↑
Apple (Quality)	Physics-MC	0.943	0.047	0.013	0.785
	Regression	0.676	0.037	0.021	0.828
	VAE	0.839	0.077	0.084	0.795
	Physics-VAE	0.494	0.024	0.010	0.867
	Real (ceiling)	–	–	–	0.886
Banana (Quality)	Physics-MC	0.350	0.086	0.076	0.883
	Regression	0.345	0.074	0.039	0.827
	VAE	0.362	0.106	0.063	0.887
	Physics-VAE	0.292	0.054	0.024	0.917
	Real (ceiling)	–	–	–	0.940
Date (Nutrients)	Physics-MC	1.129	0.208	0.083	–
	Regression	1.192	0.162	0.077	–
	VAE	0.781	0.325	0.143	–
	Physics-VAE	0.540	0.144	0.070	–
Safou (Nutrients)	Physics-MC	0.351	0.249	0.123	–
	Regression	0.653	0.301	0.130	–
	VAE	3.441	0.596	0.311	–
	Physics-VAE	0.112	0.290	0.128	–

2. Support-aware Uncertainty Evaluation

Beyond synthetic data quality evaluation, we investigated whether the uncertainty estimates produced by the conditional VAE ensemble reflect the available training support.

For this analysis, one conditional VAE ensemble was trained per region and evaluated under two transfer conditions. NEAR corresponds to held-out samples from the same region used during training, whereas FAR corresponds to samples from a different region. A desirable uncertainty model should increase predictive disagreement when operating outside the observed data distribution.

Table 2 summarizes the uncertainty behaviour. The increase in disagreement for FAR samples indicates that the ensemble captures additional uncertainty under distribution shift.

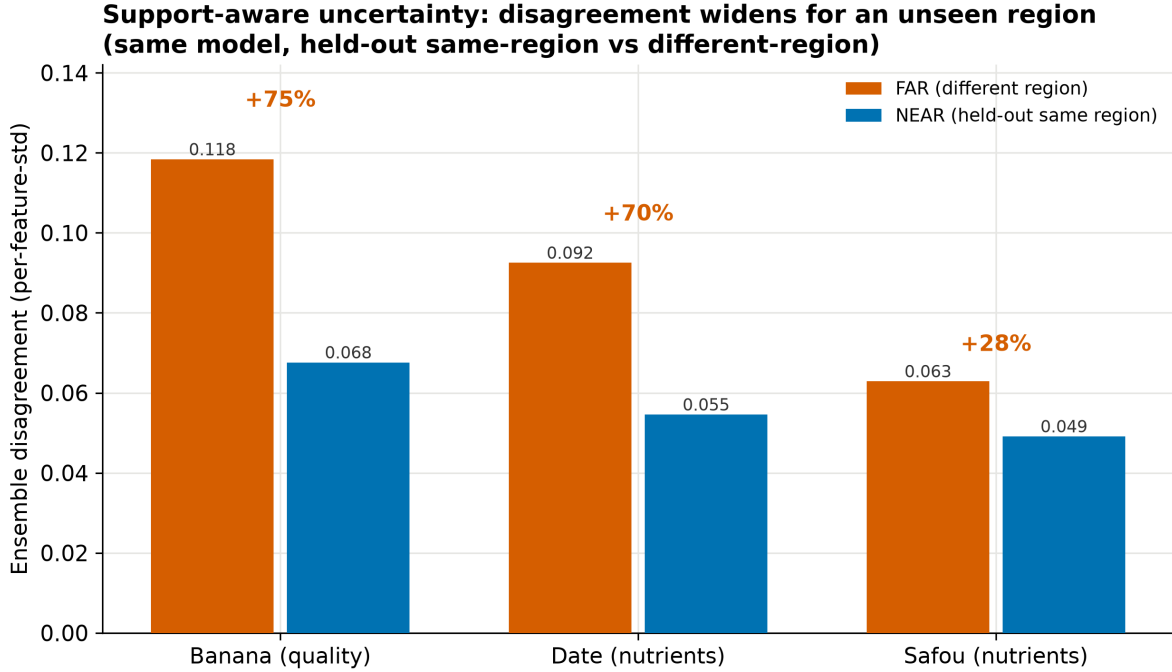


Figure 1: Support-aware uncertainty behaviour under NEAR and FAR transfer. Higher disagreement under FAR transfer indicates increased uncertainty when generating samples outside the observed training support.

Table 2: Support-aware uncertainty evaluation using conditional VAE ensembles. FAR represents transfer to a different region, while NEAR represents evaluation within the same region. Coverage@90 indicates the fraction of real observations contained within the predicted 90% uncertainty interval.

Dataset	Disagreement			Coverage@90	
	NEAR	FAR	Increase	NEAR	FAR
Banana (Quality)	0.068	0.118	+75%	0.997	0.986
Date (Nutrients)	0.055	0.092	+70%	0.696	0.737
Safou (Nutrients)	0.049	0.063	+28%	0.589	0.315